Measuring & marking out

National 5 · Drawings, measuring & marking out · Worksheet

WOODWORK WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ measure accurately with a rule, tape and calipers
- ☐ mark out with a try square, mitre square, sliding bevel, marking, cutting and mortise gauges and a marking knife
- ☐ work from a face side and face edge
- ☐ explain the advantages of templates
- ☐ work to ±1 mm when marking out and cutting

I DO — Worked example

Read how a full-mark answer is built

A class is making 20 identical curved brackets for a shelf.

1 Explain two advantages of using a template to mark out the brackets. (2 marks)

Full-mark answer each ✓ is one mark

Every bracket will be exactly the same shape, because they are all drawn round the same template ✓.
It is quicker, because the curve only has to be measured and drawn once rather than 20 times ✓.

Why marks are not awarded

"It is quicker and easier."

0 marks: no reasons given for "quicker" or "easier".

WE DO — Guided practice

Use the starters and word bank to help you

A pupil needs to mark a line across the grain at the shoulder of a tenon.

2 Name a tool that would mark the line accurately without tearing the fibres. (1 mark)

Sentence starters

1. …

Word bank

marking knife · try square · Cutting gauge

Tip: 1 marks = 1 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which tool marks a line across the grain without tearing the fibres?
☐ Marking gauge ☐ Cutting gauge ☐ Mortise gauge ☐ Pencil

b) What is the tolerance for marking out and cutting?
☐ ±0.5 mm ☐ ±1 mm ☐ ±3 mm ☐ ±5 mm

c) Which tool marks both sides of a mortise at once?
☐ Mortise gauge ☐ Marking knife ☐ Try square ☐ Steel rule

d) Why mark out the two stiles of a frame together?
☐ It makes the frame look neater ☐ It saves time sharpening pencils ☐ It is a workshop rule for frames ☐ The joints line up exactly

4 Key terms: write the letter of the matching description next to each term.

1. **Marking gauge**	____	**A** Two spurs mark both sides of a mortise or tenon at once
2. **Cutting gauge**	____	**B** Your reference surfaces: always measure and square from them
3. **Mortise gauge**	____	**C** Can be set to any angle
4. **Sliding bevel**	____	**D** Fixed at 45° for marking mitres
5. **Mitre square**	____	**E** Has a blade to cut a line across the grain without tearing
6. **Face side and face edge**	____	**F** Scribes a line parallel to an edge, along the grain

A steel rule is too short to measure the 1.2 m length of a bench top.

5 Name an alternative measuring tool. (1 mark)

A pupil is marking out a mortise in a table leg.

6 Name the tool that marks both sides of the mortise at the same time. (1 mark)

A picture frame has mitred corners.

7 Name a tool used to mark out the mitres and state the angle. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?